



MINING DEALER BEST PRACTICE

CAT 988H - Cooling System Temperature Measurement Optimization

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

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1.0 Introduction

During a normal cooling system efficiency diagnosis, the measurement of the radiator delta is requested (the difference in the inlet and outlet temperature.) This allows us to identify any abnormalities in the heat exchange system. When the result is different from the specification, there are several possibilities such as: water pump, thermostatic valve, coolers and radiator.

When troubleshooting options converge to the radiator, we must determine which radiator core is causing the problem, which can be an internal or external obstruction.

This best practice was designed to optimize machine downtime in corrective maintenance and increase the accuracy of the diagnosis by quickly identifying which radiator core has a problem, in addition to reducing maintenance costs for the customer by avoiding the replacement of unnecessary parts.

2.0 Best Practice Description

The replacement of all radiator cores is both expensive and timeconsuming. If replacing all cores, equipment can be stopped for up to 24 hours, depending on the resource availability. This Best Practice covers the process to test the cores individually in order to identify and replace only the one that has problems.

If diagnosis has found the water pump and thermostatic valves are efficient and not the cause of the issue, the individual cores should be tested. During testing, a uniform temperature between the individual cores is expected, however, if any of the cores present an irregular temperature, it is an indication that there is something abnormal and the core should be changed.

For diagnosis of the machine's cooling system, the literature REHS8874 – i05505126 highlights that the difference between the lower and upper parts of the radiator must be $4 + 5 - 5^{\circ}\text{C}$ ($39,2 + 41 - 41^{\circ}\text{F}$), as shown below.

Cooling System

Table 2

Status ⁽¹⁾	Description	Specification	Actual	Comments
	Radiator Pressure Relief (Radiator Cap) ⁽²⁾	-		
	Freezing Point of Engine Coolant	-		
	Conditioner Test of Engine Coolant	-		
	Belt Tension	-		Refer to Operation and Maintenance Manual.
	Difference between the Upper and Lower Radiator Tank Temperatures	$4 + 5 - 5^{\circ}\text{C}$ ($39 + 41 - 41^{\circ}\text{F}$)		
	Jacket Water Inlet Temperature from Cooling System	$90^{\circ} \pm 3^{\circ}\text{C}$ ($194^{\circ} \pm 5^{\circ}\text{F}$)		

(1) N = Normal, M = Monitor, A = Action

(2) Test for the pressure that is listed on the cap.

Image 01: literature for diagnostic guidance

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As the example below, where it was verified that a core had its temperature lower than the others, so it is understood that the cooling liquid was not flowing through it and that there was an obstruction.

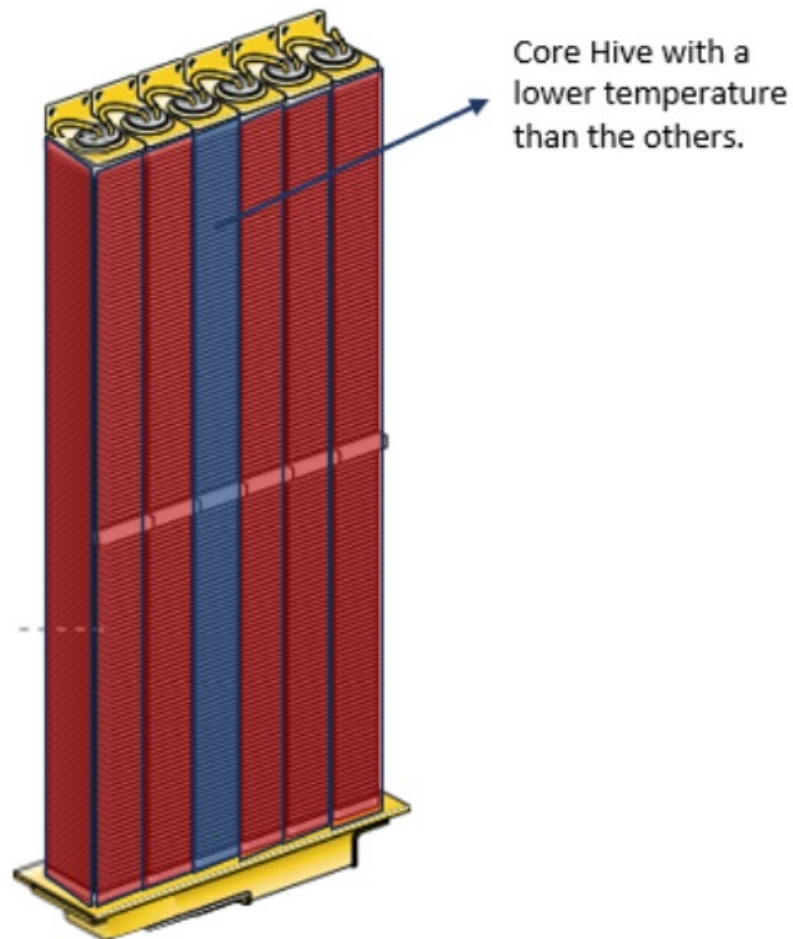


Image 2: Individual temperature measurement in each hive.

The points where the temperature should be measured are at the top of the cores near the connecting hoses between them, as shown in the images below.



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Image 3 à 8: Measurements taken at the top of the cores.

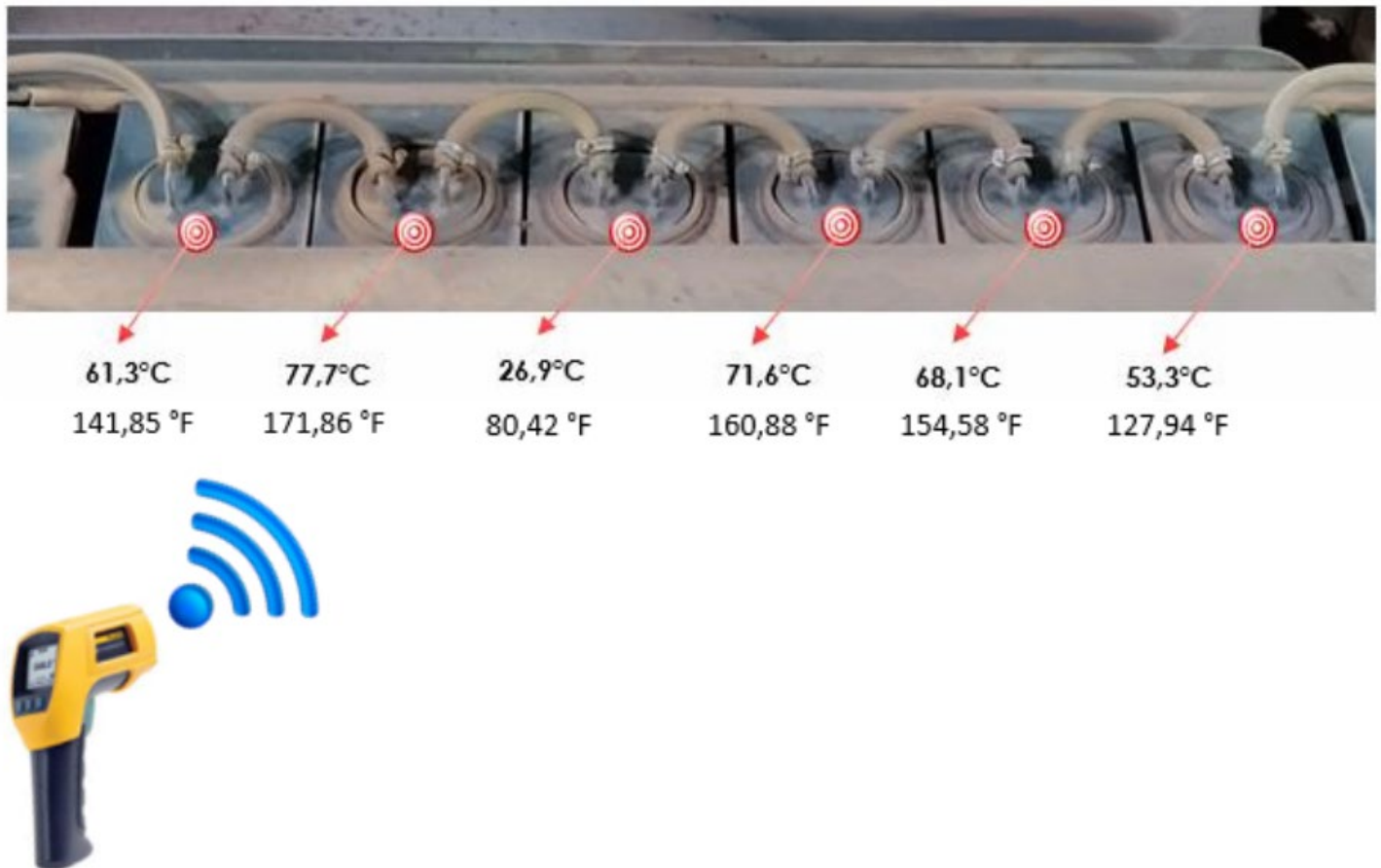


Image 9: Result of individual measurement in hives.

3.0 Implementation Steps

1. Implement the procedure described in this Best Practice into the maintenance process.
2. Establish a training and development schedule for all maintenance personnel.
3. Monitor and plan incremental improvements to the process as needed.

4.0 Benefits

- Assertiveness in the diagnosis.
- Reduced maintenance cost with trouble-free parts replacement.
- Reduced machines downtime.

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5.0 Resources Required

- Infrared Thermometer 251-0030



Image 10: Caterpillar Infrared Thermometer 251-0030

6.0 Supporting Attachments / References

Coolant Temperature Is High (REN5033-36) – i05462668.

7.0 Related Best Practices

None.

8.0 Acknowledgements

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